

Geometry Mains Olympiad (GMO)

2024 Edition

Contributors

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For supporting the idea and for invaluable help, I am especially thankful to Yousif Wameedh Ghassan Algharani (Iraq).

Finally, I, Ismayil Ismayilzada (Azerbaijan), am very thankful to myself for coming up with such a beautiful idea that showcases the potential of working with diverse communities.

Exam Rules

1. You are given 6 hours in 1 day to solve 6 problems.
2. Cheating is not allowed since we want to see people enjoy our problems.
3. You are given +10 minutes to scan your papers and send them to the provided email. (It is recommended to use Camscanner or any other scanning app.) The papers that send in 21 th october will not accepted.
4. Email address: `geometrymainsolympiad@outlook.com`
5. No camera or zoom setup is needed during the contest.
6. Have fun solving the problems!

Problems

1. Let ABC be an acute triangle. Define I as its incenter. Let D and E be the incircle's tangent points to AC and AB, respectively. Let M be the midpoint of BC. Let G be the intersection point of a perpendicular line passing through M and DE. Line AM intersects the circumcircle of $\triangle ABC$ at H. The circumcircle of $\triangle AGH$ intersects line GM at J. Prove that quadrilateral $BGCJ$ is cyclic.

2. Given a triangle ABC with hypotenuse AB ($\angle C = 90$). The excircles opposite to vertices A and B touch sides BC and AC at points T_A and T_B, respectively. Let C' be the reflection of point C about line T_AT_B. Show that the angle $\angle AC'B = 135°$.

3. Let ABC be a scalene triangle. D is the foot of the altitude from A to BC. H is the orthocenter, M is the midpoint of BC, and N is the midpoint of AH. Tangents from B and C to the circumcircle of ABC intersect at T. Prove that line TH passes through one of the intersections of circles ADT with MHN.

4. Given a triangle $\triangle ABC$ with circumcircle Ω and a circle γ passing through A and being concentric to the incircle of $\triangle ABC$. Let γ intersect AB, AC, BC, and Ω at points T_1, T_2, T_3, T_4, T_5, where $T_1 \in \Omega$, $T_2 \in AB$, $T_3 \in AC$, $T_4, T_5 \in BC$. Show that if X is the intersection of the diagonals of quadrilateral $T_2T_3T_4T_5$, then $\angle AT_1X = 90°$.

5. Given a triangle ABC with bisectors BL_2 and CL_3, and a point X that is symmetrical to L_3 with respect to L_2, show that if $\frac{p}{AL_2} = \frac{BC}{BL_3}$, where $2p$ is the perimeter of triangle ABC, then $\angle ACX = \frac{1}{2}\angle ABC$.

6. Given a triangle $\triangle ABC$ with circumcircle Ω, excircle Γ that is tangent to segment BC at T. The A-mixtilinear circle ω of $\triangle ABC$. D, E are points where ω is tangent to AB, AC. Suppose N is the midpoint of the arc BAC, and $G \in AT \cap \Omega$. Show that if the circles $(NDE), \omega$ have the same radius, then tangents to Ω at N, G intersect on DE.